Problem Chosen	2026 HiMCM Summary Sheet	Team Control Number
A		18213

Choosing a City for High-School Life with Incomplete Evidence

Summary

We compare sixteen cities for high-school life across ten indicators. Fixed-anchor TOPSIS and maximum regret recommend Zurich under baseline and equal weights; all sixteen remain possible champions within the retained evidence intervals.

Official records supplement 25 originally missing city–indicator entries without replacing team values. The mean score-interval width falls from 18.81077 to 15.95582, a 15.18% reduction. The interval revision retains 48 uncertain entries and changes 13 fixed-weight ranks from midpoint completion; separate interior completions let each city lead.

Withholding Zurich's homicide point selects Singapore under baseline weights and Oslo under equal weights. Withdrawing its work-commute proxy selects Singapore under baseline weights and retains Zurich under equal weights. Doubling the education emphasis selects Singapore. In 10,000 baseline weight perturbations at Dirichlet concentration 100, Zurich is recommended in 43.18% of draws, compared with 18.01% for Oslo and 38.41% for Singapore. These frequencies are conditional on the chosen weight distribution.

Under uniform ten-per-cent development changes, Singapore is selected under baseline weights for resource expansion and living pressure; balanced improvement retains Zurich. City-specific shocks are illustrative, not forecasts. Country, district and age-group cases distinguish facility supply from student access and broad observed groups from unsupported age-specific preferences.

Keywords: high-school opportunities; TOPSIS; interval evidence; maximum regret; preference sensitivity.

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1 Introduction and decision scope

1.1 What makes a city suitable for high-school life?

A useful comparison connects a student's school day with the opportunities and constraints around it. Safety and air quality affect the daily setting; teaching resources and nearby higher education describe educational supply; housing, living costs, transport, travel, broadband and swimming facilities describe household pressures and everyday resources. Our operational definition of "best" is the conservative choice under these ten criteria, declared weights and retained evidence ranges. It provides a starting recommendation for the compared cities, rather than a claim about every student's experience.

The study uses the sixteen cities in the frozen team matrix: Amsterdam, Berlin, Dublin, Helsinki, London, Madrid, Milan, Oslo, Paris, Prague, Singapore, Stockholm, Tokyo, Vienna, Warsaw and Zurich. This is an inherited comparison set across sixteen countries, not a representative sample of all world cities. It exceeds the problem's minimum of ten cities and spans different resource and cost settings. The findings concern this set; a broader claim would require an explicit sampling design and additional comparable observations.

1.2 Criterion selection and contribution

The ten indicators preserve the agreed team framework and balance education with the conditions that make resources usable in daily life. Their directions and anchors appear in Table 1. Counts and city averages serve as scoped supply or setting measures. They do not directly measure friendships, student wellbeing, school quality, language fit, discrimination, cultural participation or individual access. Those outcomes are relevant to a family's choice; comparable student-level observations are needed to integrate them without inventing measurements or double-counting existing proxies.

Our central contribution is a city decision that retains incomplete evidence. We improve source precision, compare a point-completion benchmark with an interval revision, and examine preferences and future conditions without changing the primary indicator set. TOPSIS provides one main quality score; AHP structures stated priorities, Monte Carlo measures specified perturbations, and VIKOR supplies a focused method comparison. The extensions test the recommendation rather than vote among an expanding set of ranking models.

AI assistance supported research retrieval, calculations, code and prose preparation, including Exa source retrieval and extraction [18, 19]. The appended use report records the known assistance and the outstanding documentation required before submission.

2 Assumptions and their role

1. **Comparability follows the retained scope.** Original values and population denominators remain fixed. Each supplement keeps its provider, year, city boundary and educational population. Mixed-year densities are conditional comparisons; they do not create a same-year census.

2. **Utilities are monotone on fixed scales.** More resources and less cost or burden improve their respective utilities. Declared anchors set the scale; they are neither estimated preferences nor empirical optimum values.

3. **The score permits compensation.** Weighted distances combine criteria in a common utility space. The model therefore allows stronger results on some criteria to offset weaker results on others. A family requiring a hard safety or eligibility threshold should apply it separately before choosing.

4. **The evidence set is rectangular.** Uncertain utilities vary independently within retained intervals. This conservative closure omits unknown cross-criterion and cross-city restrictions. Its possible champions and regret bounds are conditional on this set; they are not population probabilities.

5. **Extensions hold the comparison basis fixed.** Scenario changes retain population denominators, anchors and the indicated weight rule. They are conditional experiments with no fitted future date or representative age-specific preference distribution.

3 Notation

Symbol	Meaning
M, N	Sixteen cities and ten indicators
w_j, a_j	Normalized criterion weight and fixed utility anchor
u_{ij}	Utility of indicator j for city i, on $[0, 1]$
ℓ_{ij}, h_{ij}	Retained lower and upper utility bounds
L_i, U_i	Lower and upper TOPSIS score, on $[0, 100]$
$R_i, \mathcal{C}$	Maximum regret and set of possible champions
κ	Concentration of a declared Dirichlet weight distribution
r	Raw-value multiplier in a development scenario

4 Data collection and comparability

4.1 A traceable city–indicator matrix

We compare $M = 16$ cities on $N = 10$ indicators covering safety, environment, household costs, education and everyday resources. The initial matrix is the team-supplied city table [1]. We retain its existing values, populations, reference years and source flags as the agreed input, then supplement only fields empty in that original table. Some supplied values lack original source URLs, so their provenance remains the team table; official supplements do not retrospectively validate every supplied measurement. Table 1 states what each variable measures.

The nominal reference year is 2025; supplements retain their actual years and supplied population denominators. Matching Eurostat records give Dublin's 28.8 minutes for 2011 and Zurich's 39.9 for 2024 as journeys to work, currently labelled "greater city"; Zurich's

Table 1: Retained indicators, units and reference anchors. Supplied values are retained; supplements have explicit source scope.

Indicator	Measurement and interpretation	Anchor
Homicide	Deaths per 100,000; cost; supplied city proxy	2
$PM_{2.5}$	$\mu g/m^3$; cost; supplied concentration proxy	15
Living costs	Supplied living-cost index; cost	100
Housing	Supplied price-to-income ratio; cost	10
School resources	Pupils per teacher or explicitly scoped FTE; cost	15
Higher education	Institutions per 100,000; benefit; supply proxy	1
Transport stops	Unique stops per 100,000; benefit; partial inventories	50
Travel	Duration in minutes; $\exp(-t/30)$; commute or stage proxy	30 min
Broadband	Supplied Mbps; benefit; city/general-use proxy	100
Swimming pools	Venues per 100,000; benefit; supply proxy	1

value is flagged estimated [55]. These matches establish neither the team's source lineage nor the historical boundary. We retain the supplied means, years and flags as conditional general-commute proxies. They do not identify high-school journeys. Mixed-year densities and historical inventories remain conditional; withdrawal scenarios examine their role in the decision.

We use official statistical tables, operator inventories and municipal geodata to fill missing fields. For each admitted supplement, the archive records the provider, URL, year, geographic and population scope, units, extraction rule and file hash. Original PDFs and API responses are retained separately from derived tables. Search summaries locate evidence; numerical admission requires the underlying record or an explicitly identified provider capture. Values are converted to compatible units before utility calculation: for example, the supplied higher-education density per million is divided by ten to obtain the per-100,000 scale. Mode share is not converted into physical stop density.

4.2 Collection problems and their resolution

Count city facilities, not platforms or entire networks. Selecting Oslo's municipal GIS records within the city boundary yields 89 metro stations after consolidating 186 platforms; Zurich's 2025 VBZ report distinguishes 446 city stops from 235 outside the city [5, 6]. Helsinki's November 2026 operator snapshot similarly separates 19 Helsinki stations from 11 in Espoo [7]. For a verified subset count c, fixed population P and anchor a, we use $d = 10^5 c/P$ and utility $[d/(a+d), 1]$, leaving uncounted providers unknown. Madrid's May 2026 register has 293 records, of which 242 carry city code 28079. Omitting two unnamed records without lines and consolidating native station identifiers gives 191 complexes [23, 24]. Vienna illustrates why record totals alone are insufficient: 1,695 named areas include unverified depot objects, so only 99 metro areas individually matched to existing-station geodata and the city polygon are retained [28, 29].

Native identifiers, verified aliases and operator interchange maps prevent transfer stations from being counted once per line. Prague retains 57 current metro complexes; Flora's closure explains its omission from the 58 distinct complexes in the 2025 report, whereas Florenc's partial accessibility restriction does not close the whole complex [30, 31, 32]. Berlin's nine BVG U-Bahn profiles consolidate 131 zone-A appearances into 100 complexes. Zone A extends through the S-Bahn ring; the retained U-Bahn subset leaves outer stations and other modes uncounted [33]. Milan's municipal layer, distinguished from its entrance layer, selects records with both the Milano name and code 015146, yielding 104 stations/hubs after five map-supported interchange consolidations; proximity alone does not merge stations [49]. Tokyo's Oedo ticket-service directory lists 38 distinct stations in the 23 wards; machine flags are not added. September 2026 updates describe the catalogue; its inventory date is unstated [56]. Warsaw's 21 M1 and 18 M2 entries yield 38 complexes after merging Świętokrzyska. Its 2024 map distinguishes planned stations [57]; a 2025 fire report locates the lines within Warsaw [58]. Counts use Tokyo's 2026 and Warsaw's 2024 populations.

Amsterdam's 39 provider-grouped Metro locations give 35 inside both municipal 2026 and CBS 2025 boundaries; a declared 100-metre nominal boundary screen retains 32 [53, 54]. The screen defines a subset, not an accuracy guarantee. The 2026 metadata leave its inventory date unspecified. The retained 2025 population of 941,927 gives a conditional lower bound, leaving other transport supply unknown.

London's inventory mixes platforms and entrances with stations. From 272 station-level records, 62 have exact fare-zone value 1; it retains 59 stations/hubs after two conservative omissions and Paddington hub consolidation; representative coordinates lie within the ONS December 2025 London-region boundary [36, 37]. Singapore's LTA GeoJSON contributes 5,205 distinct nonempty bus-stop codes, without distance clustering [41]; Stockholm's 2025 city–police report identifies thirteen metro stations in two districts [48].

Keep inventory dates, status and venue units explicit. Milan's layer is dated 16 October 2024; Paris's 244 intra-muros metro stations come from a March 2022 atlas using RATP 2021 and Apur 2022 sources [49, 9]. These historical lower-bound proxies are not relabelled as 2025 counts; a separate withdrawal restores the Paris supplement to unknown. Vienna, Berlin and London use November 2026 inventories. All retain the supplied population denominators, including Paris's 2023, Dublin's 2022 and Singapore's 2024 values. Inventory membership does not establish continuous operation or adolescent access. Stockholm's November 2026 directory retains thirteen municipal swimming venues after excluding a renovation closure and counting its replacement once [8]. Tokyo retains five Nerima indoor venues at distinct addresses: Heiwadai's current closure excludes it, while Miharadai's announced February–May 2027 closure is not a current closure [25, 26]. Named subsets include 25 Singapore venues in the ActiveSG directory and one OCBC Aquatic Centre at 7 Stadium Drive, Singapore 397632, counted once despite its multiple basins; operator pages were captured 10 November 2026 and their inventory date is unstated. The denominator is the 2024 population of 6,036,900. Twelve Zurich venues and seventeen Milan indoor venues under the June 2025 tariff are also retained [42, 44, 43, 46, 47]. The Singapore count does not establish opening or teen access. Four

earlier pool-specific notices affect three ActiveSG venues: withholding them leaves 23 venues; also withholding OCBC leaves 22. At OCBC, competition-pool booking is for ages 16 and above, training-pool booking is for all ages, and supervisory rules apply for ages 15 and below. Individual per-entry prices are offered; normal hours are 07:00–21:30 Monday and Wednesday–Sunday. Tuesday maintenance, the 30 October–1 November 2026 event and 5–28 December competition-pool closure with limited training availability constrain access [43]. Zurich includes five distinct school pools with limited public sessions, some women/girls-only, closed during school holidays and non-school days; adult-only Kügeliloo is excluded. Venue supply does not establish adolescent access; other providers remain unknown.

Match school stages and teaching units before forming ratios. Prague's 2024/25 totals are 80,073 secondary-school pupils and 6,877.1 teacher full-time equivalents (FTE), including lower-gymnasium years. Subtracting the matching 8,748 pupils and 737.8 FTE gives 71,325 upper-stage pupils and a rounding envelope of $[6139.2, 6139.4]$ teacher FTE, including adult and non-daytime study modes [10]. Stockholm's city-school ratio covers upper-secondary years 1–3, all ownership types and the 15 October reference date. Its API value 16.42091405 is retained as $[16.35, 16.45]$: a declared precision convention, not a measurement-error confidence interval [11]. Warsaw's 2024/25 yearbook gives 119,740 pupils across six youth school routes by school location and all ownership types. Unknown youth allocation within adult-inclusive general-school FTE gives 2,490–8,144 FTE; outward one-FTE guards on four published cells widen this to 2,487–8,148 [45]. The guards encode a unit-rounding assumption. These allocation and rounding bounds describe school resources, not a resident age-14–18 population or a statistical confidence interval.

Some published denominators fail these checks. Henri-IV in Paris reports 794 lycée pupils and 153.65 institution FTE for 2025, without upper-stage allocation or complete city coverage; a 2026 post-baccalaureate directory flag cannot recover 2025 workloads [34, 35]. Dublin's school-roll payroll FTE are unpublished [27]; its November 2025 City Of Dublin ETB post-primary vocational subgroup has 411.8 payroll FTE against 376.1 allocated posts, so allocations are not a payroll ceiling [50]. Finland's national FTE combines register, salary and private-provider estimation; without a verified one-sided error envelope it is not a guaranteed Helsinki teacher upper bound [51]. All three city PTRs remain unknown.

Preserve incompatible rate units and supplied values. Amsterdam's CBS sources give ten provisional victims in 2022 and 68 across 2018–2022, neither matching the supplied numerator 340 [39]. Zurich's PKS lists one completed offence in 2024 and two in 2025; offences count criminal acts independently of victims, and the submitted rate 0.14 still lacks a verified denominator [40]. Both original entries remain, with their contribution tested in Section 5.5; we do not manufacture replacement rates from incompatible units.

Retain unobserved travellers and failed retrievals as uncertainty. Singapore's initial census-PDF endpoint returned 404; the official Parliament mirror and API supplied the matching report and table [13]. Duration-bin utility uses endpoints, no midpoints or artificial open-tail cap. Weighted enrolment estimates exceed travel coverage by 1,811

secondary, 989 post-secondary non-tertiary and 1,635 polytechnic students. These are weighted survey estimates, not complete headcounts. Uncovered mass retains $[0, 1]$ utility, and unknown stage shares use the interval convex hull. The resulting $[0.21048, 0.61884]$ is a 2020 school-stage proxy, not an identified 2025 interval for ages 14–18. Dublin's stop-download 403 yielded no catalogue-inferred count; a later official map from the 2023 plan, issued in 2024, supports nineteen existing city-centre tram stops [14]. Warsaw's recovered 2022 survey reports a rounded 27-minute school trip for a city subgroup with sample base 22, but covers residents aged 15 and above without an upper-secondary or matched 14–18 denominator [38]. Its travel utility remains unknown.

Figure 1 locates the remaining uncertainty. The 160 cells comprise 112 points, 37 partial intervals and 11 fully unknown entries: eight teacher-ratio and three travel entries. The original table had 112 points, 12 partial intervals and 36 unknowns. Evidence for twenty-five distinct city–indicator cells fills original gaps while preserving supplied measurements. Facility supply, commute proxies and school resources remain distinct from adolescent access, school travel and learning quality.

5 Establishing the baseline evaluation model

5.1 Determining indicator weights

We retain eight main indicators at weight 0.1125 each and two auxiliary facility indicators, transport stops and pools, at 0.05 each, so $\sum_j w_j = 1$. These weights express a transparent modelling choice. They are not estimated from a student survey. Equal weights provide a separate reference. TOPSIS is the primary score; VIKOR provides a focused method comparison. AHP represents explicit preference scenarios [2], and Monte Carlo measures sensitivity to declared distributions. These extensions retain the same criteria rather than introduce unrelated ranking models.

For preference multipliers $m_j > 0$, we use

$$w'_j = \frac{m_j w_j}{\sum_k w_k m_k}, \qquad A_{jk} = \frac{w'_j}{w'_k}. \tag{1}$$

The principal eigenvector of A recovers w'. With $\lambda_{\max}$ its largest eigenvalue, $CI = (\lambda_{\max} - N)/(N-1)$ and $CR = CI/RI_N$. Because this matrix is built from exact ratios, $\lambda_{\max} = N$ and $CR = 0$ up to numerical precision. This verifies internal consistency; it does not establish that these are students' actual preferences.

5.2 TOPSIS scoring calculation

We adapt TOPSIS [3] using fixed reference anchors to preserve a common scoring scale across evidence, preference and future scenarios. For benefit and cost measurements respectively,

$$u^+(x) = \frac{x}{a+x}, \qquad u^-(x) = \frac{a}{a+x}. \tag{2}$$

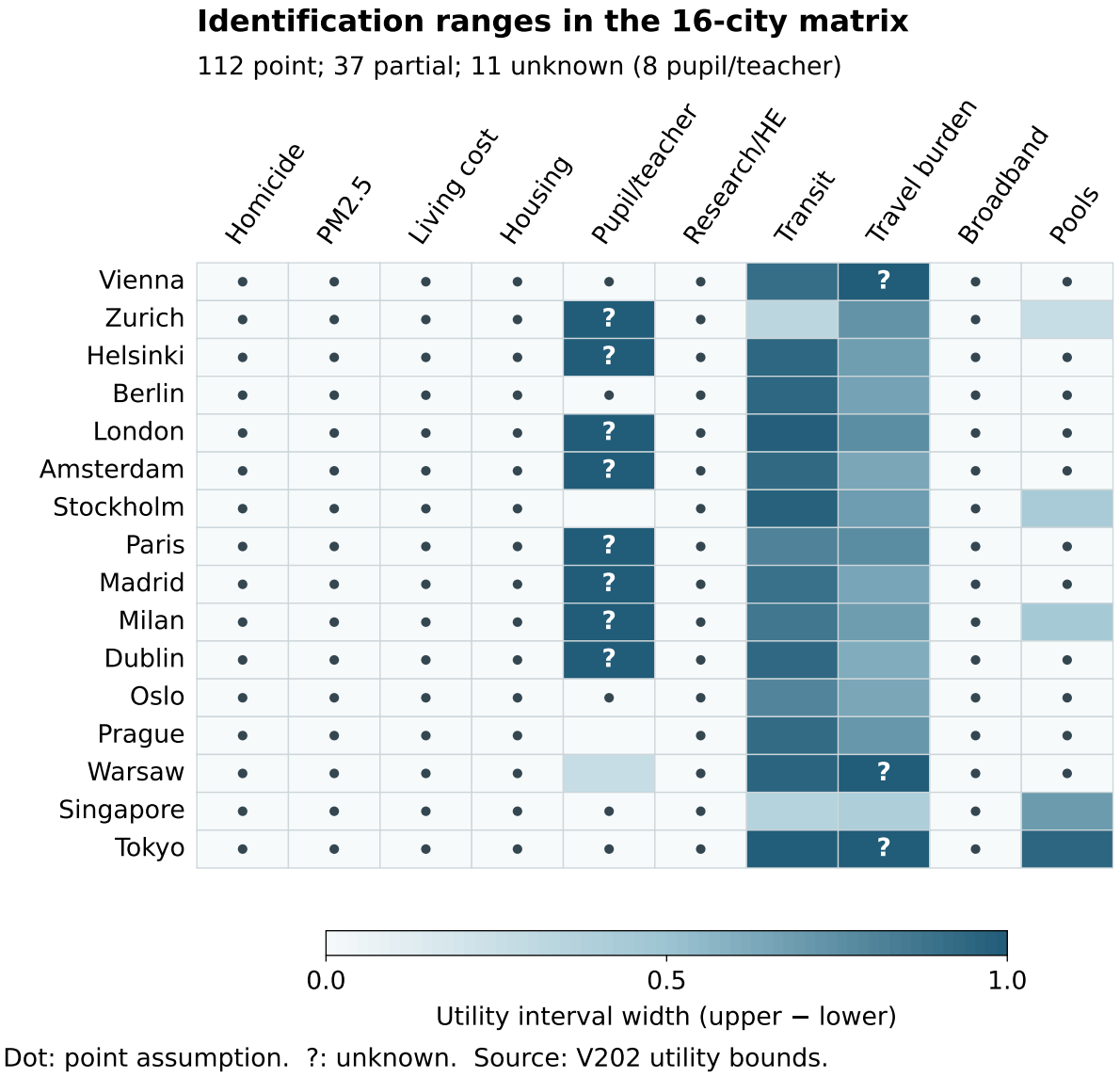


Figure 1: Identification ranges in the retained sixteen-city, ten-indicator matrix. Eight of the eleven fully unknown entries concern upper-secondary pupil–teacher ratios. Shading shows utility upper minus lower bounds; dots mark retained point assumptions and question marks identify $[0, 1]$ entries. Point assumptions include submitted values and do not certify independent validation or zero measurement error. These are identification bounds under the retained source rules, not confidence intervals. Travel entries retain school-travel and general-commute proxy scopes. Cities follow the input order. Source: V202 utility bounds.

Each anchor in Table 1 is a declared reference scale, not an estimated optimum. Travel uses $q(t) = e^{-t/30}$ for duration t in minutes. When only mean duration $\bar{t}$ is known, Jensen's inequality gives $\mathbb{E}[q(t)] \in [e^{-\bar{t}/30}, 1]$. Fully missing utilities remain $[0, 1]$.

For utility vector u_i, weighted distances from fixed worst and best points are

$$D_i^- = \sqrt{\sum_j w_j u_{ij}^2}, \qquad D_i^+ = \sqrt{\sum_j w_j (1 - u_{ij})^2}, \qquad S_i = 100 \frac{D_i^-}{D_i^- + D_i^+}. \tag{3}$$

Weights enter the squared distances once. Since S_i increases with every utility, endpoints ℓ_i, h_i give $L_i = S(\ell_i)$ and $U_i = S(h_i)$. Under the retained rectangular uncertainty set, define

$$R_i = \max\{0, \max_{k \neq i} U_k - L_i\}, \qquad i^* \in \arg\min_i R_i, \tag{4}$$

$$\mathcal{C} = \{i : U_i \geq \max_{k \neq i} L_k\}. \tag{5}$$

Maximum regret R_i measures how far the selected city could trail another city. Set $\mathcal{C}$ retains every city that could still be first. The maximum excludes the selected city itself. These conditional data bounds are not sampling confidence intervals, and a conservative recommendation does not identify a uniquely best city.

5.3 Scoring results and interpretation

Zurich minimizes maximum regret, followed by Singapore and Oslo (Table 2). Zurich's score lower bound is below those of Singapore and Oslo; its own upper bound is excluded from its competitor set. Its regret is $70.90353 - 53.96521 = 16.93832$, whereas Oslo's is $72.18379 - 54.29594 = 17.88785$. All sixteen cities remain possible champions. The decision order expresses a conservative recommendation under the stated criteria and intervals.

Table 2: Conservative baseline decision ranks with unchanged weights.

Rank	City	Score interval	Regret	Rank	City	Score interval	Regret
1	Zurich	53.97–72.18	16.94	9	Paris	49.11–69.04	23.08
2	Singapore	55.23–63.53	16.95	10	Madrid	49.00–68.22	23.18
3	Oslo	54.30–63.90	17.89	11	Warsaw	47.64–63.99	24.54
4	Stockholm	52.12–64.54	20.06	12	Vienna	47.52–61.03	24.66
5	Prague	51.97–62.78	20.21	13	Milan	46.07–65.99	26.11
6	Dublin	51.13–70.90	21.05	14	London	45.22–66.12	26.97
7	Berlin	50.73–61.54	21.45	15	Tokyo	43.98–61.12	28.20
8	Helsinki	49.41–69.84	22.78	16	Amsterdam	42.48–60.43	29.70

Holding weights and formulas fixed, evidence for twenty-five distinct city–indicator cells narrows the average fixed-weight score interval from 18.81077 to 15.95582, a 15.18% reduction. Adding OCBC Aquatic Centre to Singapore's 25-venue inventory narrows the preceding fixed-weight mean from 15.95813 to 15.95582; inventory does not establish current opening or teen access. Earlier additions improved Stockholm's decision position.

Withdrawal of the historical Paris and Dublin transport inventories still recommends Zurich; withdrawal of Singapore's historical school-stage travel proxy also preserves this recommendation. Evidence improves precision and changes decisions before it necessarily reduces the possible-champion set.

As a method diagnostic, TOPSIS and VIKOR use the same illustrative midpoint utilities and weights. VIKOR balances total and largest weighted shortfalls equally and uses sample best/worst values [4]. TOPSIS selects Zurich at the midpoint, whereas VIKOR selects Dublin; Kendall's $\tau_b = 0.500$. Dublin satisfies VIKOR's advantage and stability conditions. Under equal weights, $\tau_b = 0.633$; Oslo has the lowest VIKOR index but fails those acceptance conditions, so it is not an accepted unique compromise solution. The midpoint comparison does not replace the interval decision in Equation 4. Agreement indicates similarity within the chosen conventions; disagreement exposes method dependence rather than justifying a post-hoc choice of the preferred winner.

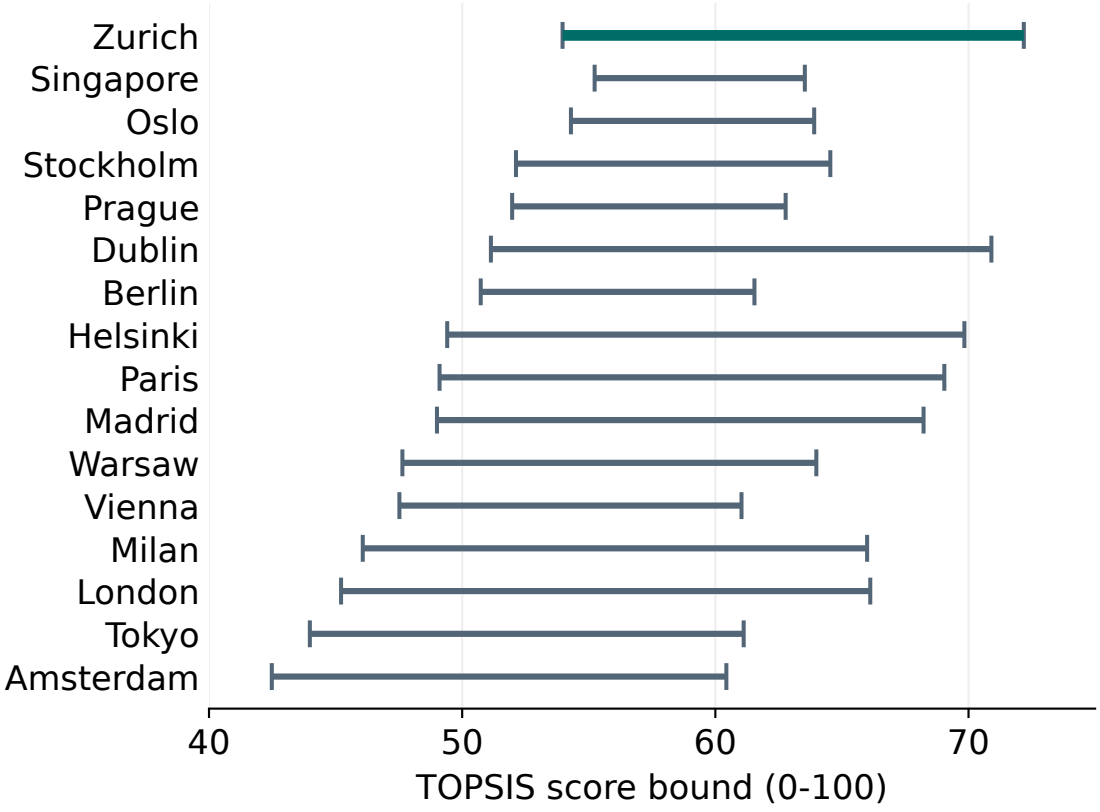


Figure 2: Current score bounds and conservative decision order for the sixteen cities. Intervals retain missing information; they are not confidence intervals. Source: frozen city matrix and admitted supplements.

5.4 A significant revision: retaining uncertain evidence

A controlled comparison shows why evidence bounds belong in the decision. On the same 160 utility entries, compare the midpoint completion $z_{ij} = (\ell_{ij} + h_{ij})/2$ with the retained-interval formulation in Equation 4. Both use identical weights, anchors and TOPSIS scores. The midpoint is a declared diagnostic scenario, rather than an estimate of missing observations or a reconstruction of the team's historical first model.

The revision retains 48 non-point entries and selects the city with the smallest worst-case shortfall. Zurich remains the recommendation, but 13 cities change decision rank under fixed weights; Kendall's $\tau_b = 0.600$. With equal weights, 14 ranks change and $\tau_b = 0.650$.

For each target city c, construct a separate interior completion: set every uncertain utility to $\ell_{ij} + \alpha_i(h_{ij} - \ell_{ij})$, with $\alpha_c = 0.95$ and $\alpha_i = 0.05$ for $i \neq c$. All 112 point values remain fixed and all 48 uncertain values lie strictly inside their intervals. Each target ranks first under both baseline and equal weights. Across the sixteen targets, the smallest lead is 4.153 score points under baseline weights and 4.491 under equal weights. These separate completions establish feasibility in the declared utility rectangle; they carry no estimated probabilities and do not establish joint physical attainability.

Retaining intervals changes the decision order while preserving sixteen possible champions. The 15.18% reduction in interval width comes from admitted data, separately from this formulation comparison.

5.5 Sensitivity to the accepted input information

Point withdrawals replace selected point utilities by $[0, 1]$, keeping weights, supplements and the 48 non-point bounds fixed. Test pre-2020 dates, post-2025 dates, non-`ok` flags and their union, counting overlaps once.

Table 3: Recommendation after withdrawing selected submitted point information. Width is the mean baseline-weight score range on the 0–100 scale.

Withdrawn group	Cells	Baseline	Equal	Width
Retained input	0	Zurich	Zurich	15.96
Before 2020	13	Singapore	Zurich	21.03
After 2025	64	Zurich	Zurich	48.34
Non-ok flags	28	Berlin	Paris	29.03
Union	95	Singapore	Singapore	68.67

Withdrawing 28 flagged points selects Berlin (baseline weights) or Paris (equal weights); removing 13 pre-2020 points selects Singapore or Zurich, respectively. Removing 64 post-2025 points retains Zurich under both, widening the baseline mean range from 15.96 to 48.34; the 95-point union selects Singapore under both. All sixteen remain possible champions. Dates and flags are screening criteria, not error probabilities.

Withholding Zurich's homicide point selects Singapore (baseline) or Oslo (equal); withholding Amsterdam's retains Zurich under both. Varying only Zurich's rate gives regret

ties near 0.14257 and 1.01449 per 100,000, respectively: Zurich uniquely minimizes regret below each cutoff and is excluded above. These are hypothetical rates, with other inputs fixed.

Withdrawing Zurich's commute interval, both identified proxies or all twelve supplied means selects Singapore (baseline), retaining Zurich under equal weights. Dublin alone changes neither choice; these withdrawals retain sixteen possible champions. Varying only Zurich's mean under Jensen bounds gives a baseline tie near 40.07 minutes: Zurich below, Singapore above; equal weights retain Zurich for every nonnegative mean, with other inputs fixed.

Jointly allow each of the twelve means t_i to vary independently in $[(1-\delta)t_i, (1+\delta)t_i]$, $0 \le \delta \le 1$; retain Jensen upper bounds, Singapore's stage bins, unknown travel and other entries. Fixed upper scores make each regret increase with its own mean: Zurich's largest mean and the other eleven smallest means maximize its regret minus the best rival's. The critical radii are approximately 0.42205% (baseline) and 63.44548% (equal). Below each boundary Zurich uniquely minimizes regret throughout the box; at it Singapore (baseline) or Oslo (equal) can tie, and above it a counterexample exists. These conditional ranges are not measured errors or future probabilities.

6 Model revision and extension

6.1 Considering future changes

We examine three conditional changes while holding anchors and population denominators fixed. Resource expansion raises higher-education, stop and pool densities by 10%. Living pressure raises living costs, housing cost ratios and travel durations by 10%. Balanced improvement reduces pupil–teacher ratios, $PM_{2.5}$ and travel durations by 10%. A comparable multi-year panel is not available, so these changes have no estimated annual rate or forecast date.

For raw multiplier $r > 0$, the utility transformations are

$$T_r^+(u) = \frac{ru}{1-u+ru}, \qquad T_r^-(u) = \frac{u}{u+r(1-u)}. \tag{6}$$

These increasing maps propagate raw-scale changes through benefit and cost intervals. For known mean travel duration, the new Jensen interval is $[e^{-r\bar{t}/30}, 1]$. Singapore instead has bounds $[L_q, U_q]$ on expected utility. With $q = e^{-t/30}$ and $t' = rt$, Jensen and $0 \le q \le 1$ give

$$\mathbb{E}[q^r] \in \begin{cases} [L_q, U_q^r], & 0 < r < 1, \\ [L_q, U_q], & r = 1, \\ [L_q^r, U_q], & r > 1. \end{cases} \tag{7}$$

Raising both endpoints to r would generally misrepresent an expected utility. Equation 7 is an outer moment bound and need not be reversible. Fully missing travel remains unknown. Thus the extension accommodates the different information supplied by means and by travel distributions.

Under fixed weights, resource expansion and living pressure select Singapore; balanced improvement retains Zurich. Equal weights retain Zurich in all three scenarios. With the Singapore historical stage-travel proxy withdrawn, Zurich remains the recommendation in all scenarios under both weights. All sixteen cities remain possible champions. We also solve a decision boundary: holding other cities fixed, proportionally reduce one city's pupil–teacher ratio, $PM_{2.5}$ and travel durations until its maximum regret ties that of the best other city. Singapore reaches the tie at approximately 0.297% reduction, Oslo at 10.152%, and Stockholm at 32.813%. Zurich already minimizes regret at baseline. A slightly greater reduction produces a strict recommendation switch in these calculations. These thresholds describe the model response to a specified joint change; they do not establish policy feasibility or a uniquely identified champion. Berlin's hypothetical combined-cost tie threshold is 41.450% with the current retained inputs. Scenario definitions and weights are unchanged.

For a separate future-shock simulation, each draw combines a common shock and city-specific shocks under a declared bounded distribution, with the criterion weights held fixed. Original data intervals are not randomly completed. Specifically, $C \sim U[0,1]$ is shared across cities and $I_i \sim U[0,1]$ varies by city, independently; $\delta_i = 0.05C + 0.05I_i$. The relevant raw multipliers are $1 + \delta_i$ or $1 - \delta_i$, and affected criteria within a city share that multiplier. With 10,000 draws per scenario, Zurich's fixed-weight recommendation frequencies are 8.00%, 49.64%, 67.70% for resource expansion, living pressure and balanced improvement respectively. Repeating the same shocks after withdrawing Singapore's historical stage-travel proxy gives 100.00%, 100.00%, 100.00% respectively. These paired calculations isolate dependence on that proxy while retaining the chosen shock distribution. These are conditional stress frequencies under an uncalibrated distribution. Weight perturbations below answer a separate question and are not blended silently into the future-shock frequencies.

6.2 Considering different preferences and development situations

Education priority doubles the weights of the two education indicators. The independent living scenario doubles travel and transport-stop weights. Equation 1 then normalizes all weights. Education priority changes the conservative recommendation from Zurich to Singapore, with decision-rank $\tau_b = 0.733$ relative to baseline. Independent-living priority retains Zurich and gives $\tau_b = 0.883$. All sixteen cities remain possible champions in each profile. The additional venue retains the education-priority choice of Singapore. The resulting differences show how the definition of a desirable city affects the decision. They do not establish measured differences between fourteen- and eighteen-year-old students.

6.3 Perturbing weights

For each preference centre w', we draw $w^{(b)} \sim \mathrm{Dirichlet}(\kappa w')$, with $\kappa \in \{20, 100, 500\}$ and 10,000 draws per setting. Every draw recomputes interval TOPSIS and maximum regret without filling missing values. We report mean decision rank, fractional first-place recommendation frequency and fractional top-three decision membership; ties share membership.

Table 4: Conditional recommendation frequencies (%) for selected cities at concentration 100.

Preference centre	Zurich	Oslo	Singapore
Baseline	43.18	18.01	38.41
Education priority	6.47	31.47	55.70
Independent living	58.21	9.58	31.86

At baseline concentration 100, Singapore, Zurich and Oslo have sample mean decision ranks 2.0515, 2.1651 and 2.3425 respectively; Zurich has the highest sample recommendation frequency. We retain minimum maximum regret as the predeclared decision rule; these conditional summaries describe its weight sensitivity. A low AHP consistency ratio checks calculation properties, not representative student opinion.

Numerical precision can be quantified separately from evidence uncertainty. For $n = 10{,}000$ independent, identically distributed draws within each setting, Hoeffding's inequality and a union bound over the $m = 2064$ exported means across 37 settings give a common event of probability at least 0.95 with error at most $R\sqrt{\log(2m/0.05)/(2n)}$, where the response range R is 1 for frequencies and 15 for ranks [52]. The resulting half-widths are 2.38 percentage points and 0.357 ranks; shared draws across outputs are allowed. On this event, subtracting two mean intervals bounds their difference. The simultaneous certificate resolves 4,260 of 4,440 pairwise mean-rank comparisons, leaves 180 unresolved, and certifies a unique smallest expected rank in 18 of 37 settings. The baseline top-city mean-rank ordering remains unresolved by this guarantee. These bounds concern integration under the declared simulation laws.

6.4 Within-city fairness and growth stages

Citywide facilities per resident measure supply. Student access additionally depends on fees, opening hours, transport, disability access and location; identifying its distribution requires matched students and qualifying facilities.

An auxiliary spatial view retains the residents and generic Learning POIs in Bruno et al.'s published grids [59]. In Singapore, 94.13% of retained residents live in cells with modelled mean walking time at most 30 minutes to 20 Learning POIs; Zurich's share is 99.97% (Figure 3). These grid means describe neither student trips nor access to the counted pools; they remain separate from the primary scores.

A separate 2025 Shanghai school-location case covers 16 districts, 235,542 regular senior-secondary pupils and 23,999 full-time teachers. Pooled resource utility is 0.604481 and the pupil-weighted district mean is 0.603267. The bottom 20% of pupil mass, with fractional boundary inclusion, has mean 0.580954, a gap of 0.022313. These district-homogeneous resource utilities measure neither individual teacher exposure nor resident access [15].

For districts d with pupil counts n_d and pupil–teacher ratios r_d, the pupil-weighted utility is $\bar{u} = \sum_d n_d \, 15/(15 + r_d) / \sum_d n_d$. Its difference from pooled utility reflects aggregation; school-location data establish neither resident travel access nor policy effects.

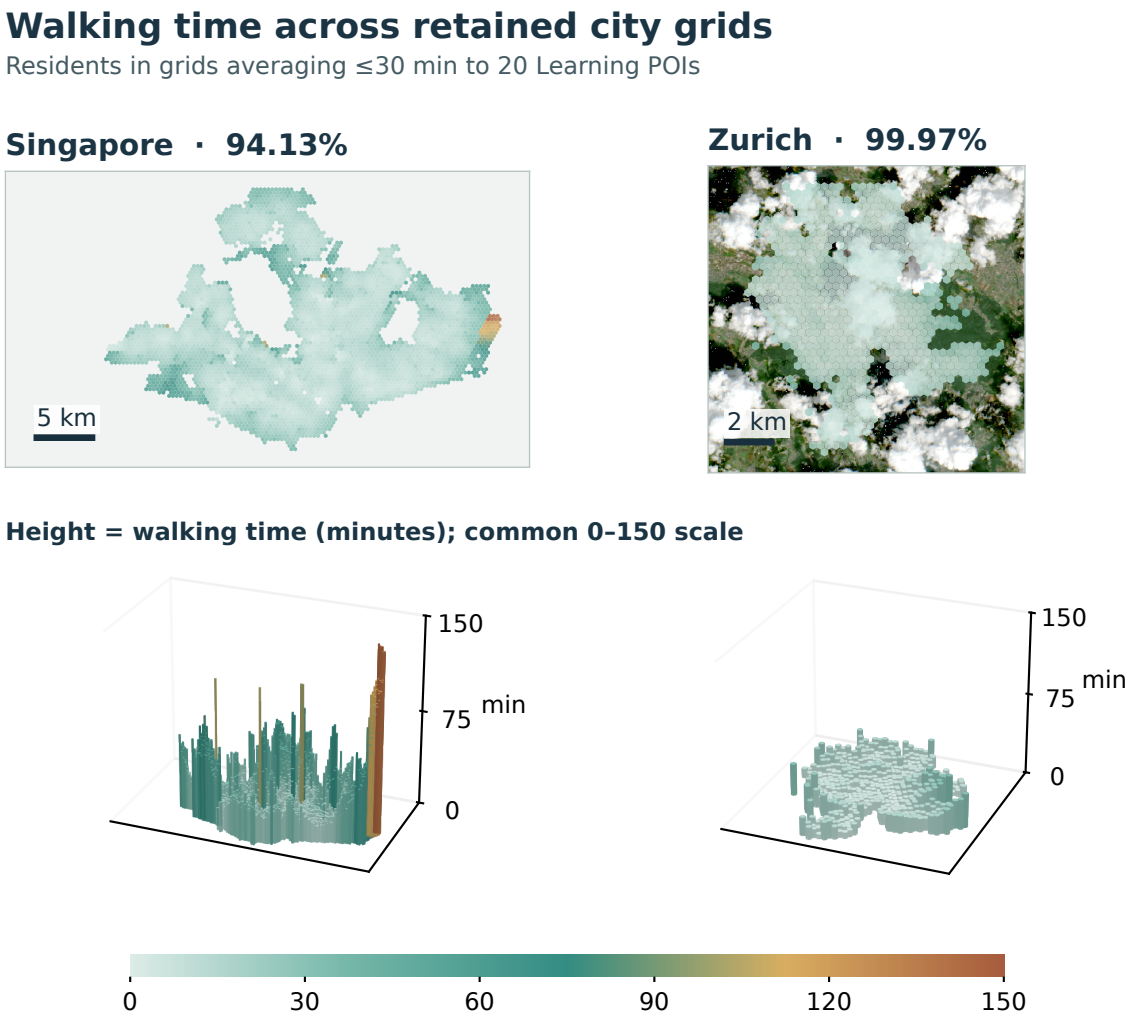


Figure 3: Modelled access to general learning facilities in retained resident grids. Colours and heights encode each cell's mean walking time to 20 Learning POIs, using a common 0–150 minute scale; height is not terrain or buildings. The labelled percentages weight qualifying cells by retained resident population. The cities have different horizontal extents, and empty cells are not a water mask. Zurich's NASA HLS S30/GIBS background was requested for 27 June 2024; cloud remains visible, and Singapore has no satellite backdrop [60]. Grids are the author-platform snapshot accessed 7 November 2026; underlying city input dates are unstated. They do not measure high-school or pool access [59].

For Istanbul's retained 2012 high-school cohort aged 14–18, let $L_{0.2}$ be the lowest-20% weighted mean travel utility, with fractional boundary inclusion, and μ the overall mean. Unknown utilities lie in $[0, 1]$; optional membership remains uncertain. Observed weight is $W = 499950$; including all optional members gives $N_{\max} = 681503$. The sharp joint bound is $\min(\mu - L_{0.2}) = (W/N_{\max})(\mu_{\text{obs}} - L_{0.2,\text{obs}}) = 0.226686$, where "obs" denotes observed trips. Including all optional members and setting unknown trips to 45 minutes attains it. With cohort, weights and missing-data rules fixed, allowing arbitrary utility changes to at most 5% or 20% of observed weighted mass leaves sharp minimum gaps 0.200134 or 0.129091, respectively. This fractional-mass relaxation uses declared error budgets, not estimated error rates or fractions of students. The official-portal aggregate projection, rather than original CSV bytes, supports these historical feasibility results, not current facility-use estimates [16].

Istanbul's 2012 household survey provides current-high-school age groups 14–15 and 16–18. Their travel-utility marginal bounds overlap: $[0.35961, 0.55711]$ and $[0.29347, 0.59346]$. Unknown durations, optional members and shared unknown-age mass mean these extrema need not be jointly attainable. They do not identify today's exact ages fourteen and eighteen. Singapore's 2020 MOE table records 38,555 fourteen-year-olds and 165 eighteen-year-olds in secondary schooling. The latter is not a resident denominator; these counts cannot join census school-stage travel bins without matched populations and reference days [16, 17].

Assigning shared unknown-age mass once gives younger-minus-older mean travel utility the joint closure $[-0.23330, 0.26325]$. Two finite-duration completions use unknown utilities 0.05 and 0.95, allocate the shared mass to the younger group, and yield contrasts $+0.23842$ and -0.20848. Closure endpoints allow zero utility as a limit; the interior reversals are feasibility constructions for the retained 2012 aggregates, not observed age effects.

Comparing exact ages fourteen and eighteen requires matched resident utilities $u_{ij}^{(14)}, u_{ij}^{(18)}$ and separately elicited weights $w_j^{(14)}, w_j^{(18)}$. Our analyst preference scenarios carry no age labels.

7 Comparing cities within the same country

Every primary city belongs to a different country, so the main sample cannot identify within-country city contrasts. We use the existing United States and India cases as frozen legacy25 V117 examples with the same ten indicator definitions and scoring formula [20]. They do not enter the sixteen-city ranking, and comparison within a country does not remove differences in source year, city boundary or educational coverage.

Phoenix has higher retained affordability utilities than New York. Mumbai has the higher $PM_{2.5}$ utility; Delhi leads in living-cost, housing, school-resource and broadband utilities. Both score pairs overlap. Affordability uses Numbeo's 2025 named-city aggregates [21]; Mumbai–Delhi air uses 2021 WHO monitoring means [22]. The retained historical city and education scopes identify neither a necessary winner nor a national causal effect.

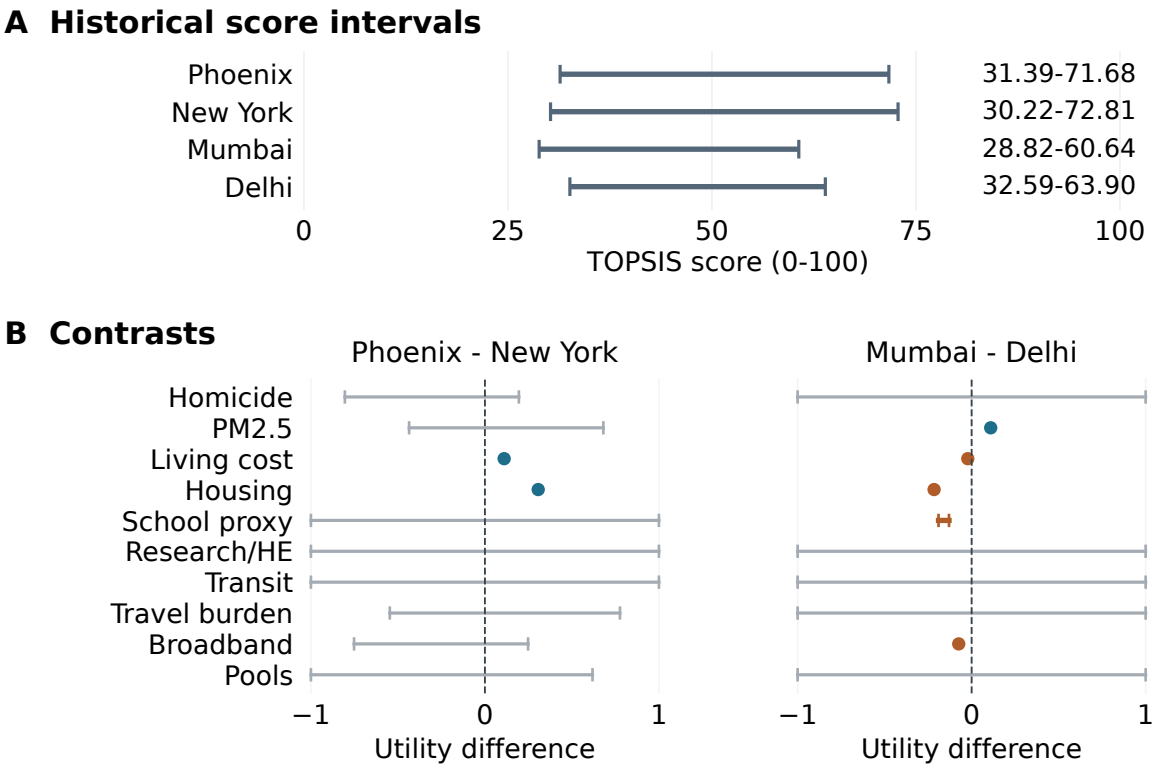


Figure 4: Frozen auxiliary V117 country cases, separate from the primary sixteen cities. Top: retained score intervals. Bottom: all twenty utility differences, first city minus second; seven exclude zero. Positive differences favour the first city; negative differences favour the second. Grey ranges leave direction unresolved. School resource retains 2024–25 school-location all-age XI–XII counts and nominal teacher-headcount allocation, not FTE. Living cost and housing use 2025 Numbeo aggregates; air uses 2021 WHO monitoring means; broadband uses the September 2026 index (July–September test window). Source years, boundaries and populations differ. Ranges are conditional identification bounds, not confidence intervals; neither pair identifies a necessary overall winner [20, 21, 22].

8 Conclusion and decision use

Zurich minimizes maximum regret under baseline and equal weights; all sixteen cities remain possible champions. Education priority selects Singapore. Evidence supplements narrow the mean score interval by 15.18%; the interval revision changes 13 decision ranks. Families can compare priorities and source bounds.

A city choice built around a student's day

Our comparison of sixteen cities across ten aspects of high-school life gives students and families a starting point for weighing everyday trade-offs. Under the retained team inputs, Zurich is the cautious recommendation; an education-focused profile points to Singapore. The priorities are analyst-designed using the Analytic Hierarchy Process (AHP), not drawn from a student survey.

The measures cover teaching resources, higher education, air quality, safety, housing and living costs, transport stops, journey time, internet access and swimming facilities. Missing or partial figures remain ranges instead of being filled with invented averages. The recommendation compares each city's largest possible shortfall against a stronger alternative within those ranges. Zurich has the smallest such shortfall under baseline priorities and when all ten measures receive equal weight. All sixteen cities remain possible leaders; the comparison does not establish one certain winner.

Clearer records, more useful choices

Additional evidence for twenty-five previously missing entries narrows the average city-score range by about fifteen per cent while preserving the team's original values and scoring rules. Source definitions matter too: Vienna's broad transport register includes depot objects not confirmed as passenger stops, while a cross-check confirms 99 metro station areas. Prague's city list gives 57 metro complexes when interchanges count once and a closed station is omitted. Records span different years, and some describe city resources rather than a teenager's experience.

Source checks show where a recommendation depends on a particular input. Treating Zurich's unverified homicide-rate figure as unknown points to Singapore under baseline priorities and Oslo under equal priorities. Setting aside Zurich's general work-commute proxy points to Singapore under baseline priorities, while equal priorities retain Zurich. That proxy does not describe high-school journeys; the supplied figures remain in the baseline.

Counts describe resource supply, not a student's access. A stop count does not show walking distance, service, fares or accessible routes; a pool count does not show fees, opening hours or neighbourhood reach. Families and planners can check these conditions alongside school options, budgets and daily routes.

Scenarios for planning

In hypothetical tests where affected quantities change uniformly by ten per cent, resource expansion and living pressure select Singapore under baseline priorities; Zurich remains the recommendation for balanced improvement and under equal priorities. Repeated simulations vary the assumed changes across cities. These are conditional model outcomes, not forecasts or estimates of a city's real chance of being best. A family can use the comparison to discuss priorities and identify local information needed for a decision that fits a student's routine.

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Report on Use of AI Tools

Documentation status: incomplete review record. This appendix records assistance supported by saved task metadata and research receipts. The complete team AI history remains to be reconciled with actual saved conversations and other tool records before submission.

Known tool and scope

OpenAI Codex, used 8–10 November 2026. The task log records Codex Desktop client version `0.166.0-alpha.5.1` and model identifier `gpt-6.1-sol` in all 74 saved root turn contexts within the inspected prefix. The identifier does not expose an immutable serving build or establish the model used by every concurrent assistant. The assistant supported official-source search and extraction, data-scope checks, Python calculations and audits, LaTeX drafting, and wording revisions. Concurrent assistant tasks prepared bounded evidence checks, a controlled model diagnostic and the news-article contribution [18]. The anti-defensive-writing skill supplied editorial instructions; it is a workflow resource, not a surveyed student preference or a data source.

Exa, used 8–9 November 2026 through the recorded `web_fetch_exa` tool, retrieved and extracted text from official webpages, PDFs and data interfaces [19]. Saved receipts identify the service, source URLs and returned text, but do not specify an internal model or service version. They do not establish whether a generative model participated in every extraction. Extracted representations are distinguished from original downloaded bytes. For example, Tokyo sports-facility PDF text with ambiguous row order was excluded; the admitted five-venue lower bound uses separately downloaded Nerima municipal HTML.

Recovered records and coverage

The supporting inventory authenticates a complete-line prefix of this root task log by byte count and SHA-256. It ends at 14:31:18.850 UTC on 9 November 2026 (22:31 in Singapore); subsequent work is outside the window. A private export retains 326 selected visible text records: six selected genuine requests, 259 progress messages and 61 complete final responses. The original 125-record selection and its private byte prefix are preserved.

A second private export retains 1,865 plaintext assistant tool-call inputs: 1,749 custom-call input strings and 116 function-call argument strings. These preserve generated code and LaTeX where present in the calls, extending the evidence beyond visible messages. Recorded inputs do not establish that each invocation succeeded or that every nested tool ran.

This capture covers one root session. It excludes tool results, attachment contents, sub-agent response sessions, other chats and other team members. Reasoning, system and developer messages, encrypted fields, internal metadata and injected continuations are excluded. Public inventories contain metadata and record hashes; the original log and exported text remain private and are not included as submission transcripts. No missing prompt or output has been reconstructed and labelled as an exact historical quotation.

Verification associated with this draft

The research calculations preserve the original city CSV and the V202 primary evidence. The V202 validation independently recomputes weighted norms and self-excluded competitor maxima, verifies interval membership of constructed completions and checks source hashes. Original provider records are kept separately from derived tables. Compilation and rendered-page review check the presentation. These checks establish the specified calculation and document properties; they do not replace the team's responsibility to review factual claims, original supplied data and final authorship.

Records required to finalize this report

The final report must identify every AI tool actually used, its available model or version, date and purpose, and associate relevant use with the text and references. Actual saved query and response records should be inserted according to the applicable COMAP policy, including its stated exceptions. Other team members' tools, translation use and earlier conversations remain to be confirmed from their records.

The current in-text disclosure and references identify Codex and Exa assistance. Submission readiness also requires a verified Team Control Number in place of the review header and confirmation that this report accounts for the team's actual use.